

Issues Analysis – Digital Library Management System

Project Overview

This project involved debugging, completing, and improving a Digital Library Management System built with JavaScript, HTML, and CSS. The application manages books, members, borrowing operations, statistics, and persistent storage using localStorage.

Issues Identified

1. DigitalBook Constructor Mismatch

Problem:

The `DigitalBook` constructor expected parameters that did not match the data being restored from storage. This caused digital books to fail when loading.

Solution:

The constructor was simplified to only include the required properties, and `Storage.js` was updated to pass the correct arguments.

2. Storage Restoration Errors

Problem:

Digital books were not recreated correctly when loading saved data, causing tests related to storage restoration to fail.

Solution:

The loading logic in `Storage.js` was corrected to instantiate `DigitalBook` objects properly and restore their download counts and borrowing information.

3. Image File Issues

Problem:

Several book cover images had incorrect filenames and extensions, resulting in broken images within the application.

Solution:

The images were renamed consistently and all references were updated to match the correct filenames.

4. HTML Accessibility Improvements

Problem:

The HTML lacked several accessibility features and semantic improvements.

Solution:

The following enhancements were implemented:

- Added a favicon.
 - Added a "Skip to main content" link.
 - Improved ARIA labels.
 - Corrected heading structure.
 - Improved semantic HTML organization.
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5. CSS Improvements

Problem:

The original stylesheet lacked responsive behavior and consistent styling.

Solution:

The stylesheet was reorganized and improved by adding:

- Responsive layouts.
 - Improved spacing.
 - Better typography.
 - Hover animations.
 - Responsive navigation.
 - Improved form styling.
 - Better card layouts for books and statistics.
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6. Digital Book Functionality

Problem:

Some methods referenced properties that no longer existed after refactoring.

Solution:

Methods such as `getInfo()` and `getDetails()` were updated to use the current class structure and remove obsolete properties.

7. Testing

Problem:

Several unit tests initially failed due to constructor mismatches and storage restoration issues.

Solution:

The implementation was corrected until all automated tests passed successfully.

Final Outcome

The completed application now:

- Successfully manages books and members.
- Supports digital books.
- Saves and restores data correctly.
- Provides a responsive user interface.
- Includes accessibility improvements.
- Passes all automated unit tests.

Test Results

- Test Suites Passed: **6 / 6**
- Tests Passed: **106 / 106**
- Failed Tests: **0**

The project is fully functional and meets the required project specifications.